XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 1



Lecture 1-5

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1. Estimate the equilibrium electron and hole concentrations in silicon for the following conditions:

   (*a*) $T = 300$ K, $N_d = 10^{15}$ cm^{-3}, $N_a = 4 \times 10^{15}$ cm^{-3}
   (*b*) $T = 300$ K, $N_d = 3 \times 10^{16}$ cm^{-3}, $N_a = 0$
   (*c*) $T = 300$ K, $N_d = N_a = 2 \times 10^{15}$ cm^{-3}
   (*d*) $T = 375$ K, $N_d = 0$, $N_a = 4 \times 10^{15}$ cm^{-3}
   (*e*) $T = 450$ K, $N_d = 10^{14}$ cm^{-3}, $N_a = 0$

2. Find the derivative with respect to energy of the Fermi–Dirac distribution function. determine the derivative with respect to energy for (*a*) $T = 0$ K, (*b*) $T = 300$ K, and (*c*) $T = 500$ K.

3. **Jin** knows about the electron and hole concentrations are the function of energy in the conduction band and valence band peak at a particular energy that can be seen in Figure A1.1. Now Jin assuming the $E_c - E_F = 0.20$ eV which is about the silicon. You need to estimate the energy, relative to the band edges, at which the concentrations have peak. (**15 marks**)

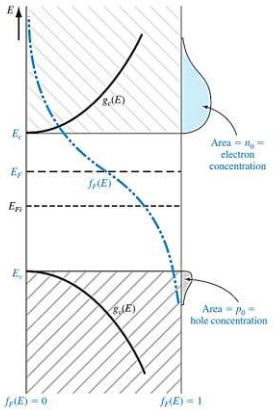


Figure A1.1

4. The lattice constant of a single crystal is 4.73 Å. Find the surface density (#/cm^2) of atoms on the (i) (100), (ii) (110), and (iii) (111) plane for a (*a*) simple cubic, (*b*) body-centered cubic, and (*c*) face-centered cubic lattice.

5. Timothy is studying about the electron by considering the infinite potential well. He wants to assume that in this case the well has energy ***V = 0*** in condition when the ***0 < x < a***.

   a) **Formulate** the wavefunctions and sketch of it by considering the **lowest three energy levels**. Find the values such as ***x***, P_x (momentum) and **E** (energy) of each these states (you need to draw all graphs).
   b) Now Timothy wants to know about the position so that he wants to use $\langle \Delta x \rangle = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$. A similar expression holds for $\langle \Delta p_x \rangle$. If the operator **for** p_x^2 **is** $-\hbar^2 \partial^2 / \partial_x^2$, check the Heisenberg uncertainty principle **($\Delta x \Delta p_x \geq \hbar/2$)** for the lowest energy state.

6. Eng Wen Ming learnt about the face-centered cubic lattice. He Assumed the atoms are hard spheres with the surfaces of the nearest neighbors touching. Also assume the effective radius of the atom is 2.37 Å. (*a*) Find the volume density of atoms in the crystal. (*b*) Find the surface density of atoms in the (110) plane. (*c*) Find the distance between nearest (110) planes. (*d*) Repeat parts (*b*) and (*c*) for the (111) plane.

7. (*a*) Estimate the total number (# / cm^3) of energy states in silicon between E_c and $E_c + 2kT$ at (*i*) $T = 300$ K and (*ii*) $T = 400$ K. (*b*) Repeat part (*a*) for GaAs.

8. Nicholas is interested about the silicon so he wanted to study about the conduction band of this material accordingly. He learnt that typically, only the states near the bottom of the conduction band can be occupied with electrons. Now he is assuming that all states within 0.1 eV of the bottom of the band that are occupied, now he needs your help to answer these questions as:
   a) He wants to know how many electrons are in the conduction band? (Use unit of cm^{-3}).
   b) Now compare this number to the atomic density of Si (use in percentage).
   (For Si, $m_n^* = 1.18 m_0 = 1.18 \times 9.11 \times 10^{-31}$ kg)

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